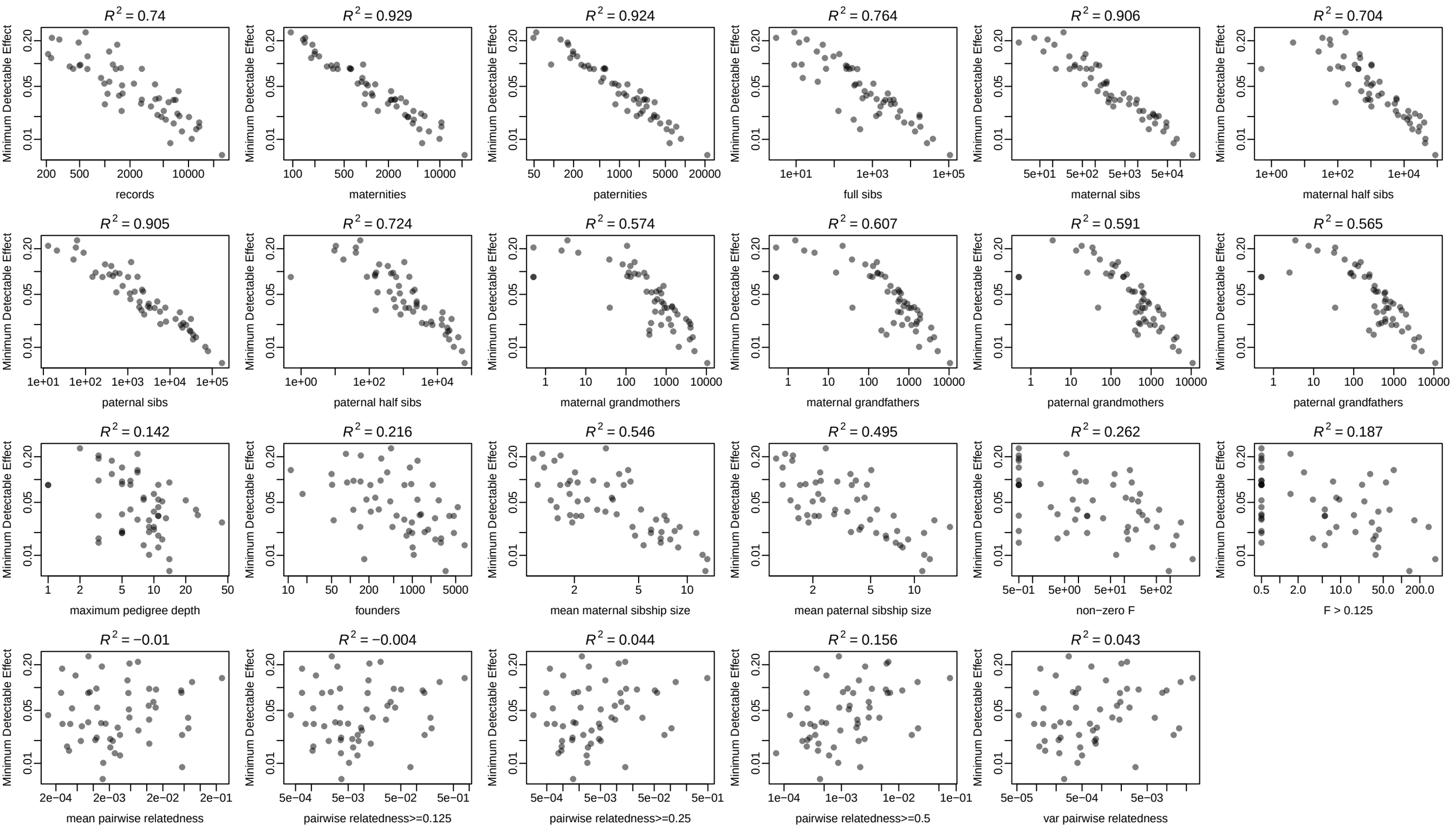

R^2 = 0.74
Minimum Detectable Effect
0.20
0.05
0.01
200 500 2000 10000
records
R^2 = 0.929
100 500 2000 10000
maternities
R^2 = 0.924
50 200 1000 5000 20000
paternities
R^2 = 0.764
1e+01 1e+03 1e+05
full sibs
R^2 = 0.906
5e+01 5e+02 5e+03 5e+04
maternal sibs
R^2 = 0.704
1e+00 1e+02 1e+04
maternal half sibs
R^2 = 0.905
1e+01 1e+02 1e+03 1e+04 1e+05
paternal sibs
R^2 = 0.724
1e+00 1e+02 1e+04
paternal half sibs
R^2 = 0.574
1 10 100 1000 10000
maternal grandmothers
R^2 = 0.607
1 10 100 1000 10000
maternal grandfathers
R^2 = 0.591
1 10 100 1000 10000
paternal grandmothers
R^2 = 0.565
1 10 100 1000 10000
paternal grandfathers
R^2 = 0.142
1 2 5 10 20 50
maximum pedigree depth
R^2 = 0.216
10 50 200 1000 5000
founders
R^2 = 0.546
2 5 10
mean maternal sibship size
R^2 = 0.495
2 5 10
mean paternal sibship size
R^2 = 0.262
5e−01 5e+00 5e+01 5e+02
non−zero F
R^2 = 0.187
0.5 2.0 10.0 50.0 200.0
F > 0.125
R^2 = −0.01
2e−04 2e−03 2e−02 2e−01
mean pairwise relatedness
R^2 = −0.004
5e−04 5e−03 5e−02 5e−01
pairwise relatedness>=0.125
R^2 = 0.044
5e−04 5e−03 5e−02 5e−01
pairwise relatedness>=0.25
R^2 = 0.156
1e−04 1e−03 1e−02 1e−01
pairwise relatedness>=0.5
R^2 = 0.043
5e−05 5e−04 5e−03
var pairwise relatedness